Bruce Shi

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SUMMARY OF QUALIFICATIONS

- Mechanical Engineering student focused on **3D CAD-driven product design, R&D prototyping,** and **manufacturing validation,** with experience across mechanical design, DFM/DFA, quality/failure analysis, and full-cycle product development
- Hands-on experience connecting CAD to fabrication through **3D printing, machining,** and **composite manufacturing**

EXPERIENCE

Knorr-Bremse | *Engineering Design Intern* May 2025 – Aug 2025

- Supported CAD-driven product development of **high-speed rail air supply units** using Creo Parametric
- Produced 300+ **technical drawings** for parts and assemblies, includes dimensional drawings, exploded views, GD&T, BOM callouts
- Designed 100+ **3D models**, including sheet metal, piping, cabling, and welded components, applying DFM/DFA principles to improve fit, manufacturability, and assembly feasibility
- Implemented 100+ engineering changes across **assemblies**, updating CAD models and drawings to support rapid design iterations
- Maintained 300+ **engineering documents** in Windchill PLM, including BOM, drawing revision, document control, and change record

Linamar | *Quality Assurance Intern* Jan 2026 – Apr 2026

- Interpreted engineering drawings and GD&T to perform dimensional validation on 1000+ automotive components using CMM
- Troubleshot 100+ non-conforming components using CMM inspection data to identify out-of-tolerance features and support corrective actions with production, quality, and engineering teams, contributing to a **5% scrap reduction**
- **Designed inspection fixtures** in SolidWorks to improve part locating, inspection repeatability, and **reduced setup time by ~30%**
- Conducted root cause analysis, linking defects to design limitations, material behavior, and manufacturing process variation to support DFM-focused process improvements
- Evaluated 100+ welded and heat-treated samples through metallurgical inspection to identify defects and material property impacts

Baja SAE Design Team – Chassis Subteam | *University of Waterloo* Sept 2025 – Present

- Developed vehicle chassis to final frame design in SolidWorks, balancing strength, weight, and manufacturability
- Supported **FEA/structural analysis** to evaluate stress, stiffness, and potential failure points under expected loading conditions
- Assisted cutting, machining, and fabrication of frame components for chassis assembly

SELECTED PROJECTS

Custom Ergonomic Carbon Fiber Mouse | *Personal Project* Jan 2026 – Apr 2026

- Developed a custom ergonomic mouse through **full-cycle product development,** converting hand geometry into a CAD model
- Reverse-engineered electronics from a Logitech M546 mouse and designed internal PCB mounts with 0.4 mm fit accuracy
- Iterated **10+ FDM 3D-printed prototypes** to validate ergonomics, component fit, and assembly feasibility
- Designed molds and fabricated a carbon fiber composite shell, **reducing weight by 20%** while improving strength

Carbon Fiber Case | *Personal Project* Jun 2025 - Aug 2025

- Designed multi-piece molds for the AirPods case in Creo with tolerance allowances to ensure proper fit for 4 carbon-fiber layers
- Used carbon fiber sheets with AB epoxy resin, applied **hand layup** and **vacuum bagging** techniques

V12 Engine CAD & Simulation | *Personal Project* May 2025 - June 2025

- Modeled 50+ components of a V12 engine in SolidWorks from scratch and assembled them with correct mates
- Conducted a kinematic simulation using Motion Study to replicate the piston-crankshaft motion and valve operation

McLaren 720S Spider – High-Fidelity 3D Modeling | *Personal Project* Jan 2024 - Jan 2025

- Modeled 100+ Blender meshes from blueprint references and produced final Cycles renders to communicate surface modeling, geometry control, and product visualization

SKILLS

CAD & Simulation:	SolidWorks, Creo, AutoCAD, CATIA, Ansys (FEA), Blender
Design & Fabrication:	GD&T, DFM/DFA, FDM/SLA 3D printing, CNC machining, laser cutting, composite manufacturing
Testing and Validation:	CMM, tolerance/fit validation, design review, BOM management, root cause analysis, failure analysis
Technical Knowledge:	heat treatment effects, weld defects, tolerance stack-up, ISO standards, PLM workflow

EDUCATION

University of Waterloo | *BASc in Honours Mechanical Engineering* Sept 2024 - April 2029

- Cumulative GPA: 89.94%
- Relevant course: Control of Material Properties, Electromechanical Devices & Power Processing